

SFA Delivery, Deployment, and Operations

Document 07 of the SFA Product Information Pack. Audience: DevOps engineers, platform operators, engineering managers, NotebookLM ingestion. **Sources:** `documentation/02-architecture/sfa-delivery-tier.md`; `documentation/05-admin-and-operations/UI_DEPLOY_RUNBOOK.md`; `documentation/05-admin-and-operations/FTPS_CRED_ROTATION_SYNC_RUNBOOK.md`; `documentation/08-troubleshooting/README.md`; `documentation/KNOWN_DEBT.md`; `documentation/08-troubleshooting/` (Docker workstation doc).

Abstract. The SFA platform is operated across three physically distinct machines with non-overlapping roles. This document covers the hosting topology and the rationale for each machine's role; the two-part deploy pipeline (code via FTPS relay, data via HMAC ingest); cron automation and freshness guards; the complete security model (HMAC, IP allowlist, FTPS explicit TLS, migration token gate); Docker port canon for the shared development workstation; and a high-level summary of known operational debt. Operators and engineers planning new features or infrastructure changes should read this document before touching anything deploy/host/network related.

1. Three-Host Topology

The word "host" refers to three completely different machines. Never conflate them.

Role	Machine	What it does	What it does NOT do
Web host (the live site + MySQL)	uPress shared LAMP <code>sfa.nimrod.bio</code> , Cloudflare edge	Serves ALL end-user HTTP; hosts the live MySQL read-mirror; terminates TLS via Cloudflare	—
Backend / pipeline host	waldhomeserver (home server, private network, Tailscale)	Canonical Postgres SSoT; all scrapers, normalizer, aggregator, cron; AOS infra agents	NEVER serves end-user HTTP; not a staging mirror
Deploy relay / data push origin	waldhomeserver (egress IP allowlisted by uPress on port 21) for <i>code</i> ; Mac or waldhomeserver for <i>data</i>	Runs <code>lftp mirror</code> (code FTPS) and/or <code>sfa_ingest_push.py</code> (data HTTPS) toward uPress	Does not host or serve anything to end users

Why this topology exists

Three separate concerns drove the three-machine split:

1. **End-user HTTP needs a reliable public host with a proper SSL cert.** uPress provides this: shared LAMP on a CDN-backed hostname (`sfa.nimrod.bio`), Cloudflare Universal SSL, nginx managed by uPress. Running a web server on the home server would require a static IP, port forwarding, cert management, and DDoS exposure — not viable for a shared-workstation home network.

2. **Long-running background jobs (scraping, Playwright, cron) need a persistent always-on machine that can't run on shared hosting.** uPress shared hosting caps CPU, has no persistent processes, no cron that runs long jobs, and no Playwright/headless Chromium. waldhomeserver fills this role.
3. **FTPS deploy to uPress requires an allowlisted egress IP.** uPress restricts FTPS port-21 access to a short IP allowlist. The Mac's Bezeq residential ISP IP is not on the allowlist (dynamic IP, not registered). waldhomeserver has a static IP (46.235.231.114) registered on the allowlist. HTTPS (port 443) to Cloudflare has no such restriction — so data pushes can originate from either the Mac or the server.

2. Deploy Pipeline

There are two distinct deployment operations with different transports, different origins, and different cadences.

2.1 Code Deploy (UI, CSS, JS, PHP)

What is deployed: the `sfa_delivery/` directory — PHP/Slim app, Composer vendor, CSS/JS assets, templates, migrations.

Transport: FTPS port 21 (explicit TLS, `prot_c` required), lftp mirror with `--delete`.

Origin: waldhomeserver ONLY (Mac's Bezeq IP is port-21-blocked by uPress firewall).

Script: `scripts/ftp_deploy_sfa_ui.sh`

Steps the script performs: 1. Load `.env` (reads `SFA_FTP_HOST`, `SFA_FTP_PORT=21`, `SFA_FTP_USER`, `SFA_FTP_PASS`, `SFA_FTP_ROOT=/`). 2. Run `composer install --no-dev --optimize-autoloader` in `sfa_delivery/` to produce the production `vendor/` directory (gitignored; never in the repo). 3. Verify `vendor/` exists (guard against silent Composer failure). 4. Run `lftp mirror -R --delete` from the local `sfa_delivery/` tree to `SFA_FTP_ROOT` on uPress, excluding: `.env*`, `.git*`, `logs/`, `tests/`, `.DS_Store`, `*.pyc`, `__pycache__`.

The `--delete` flag makes the remote tree mirror the local source exactly. This means re-deploying an earlier commit is a rollback: the remote will match the local source, no manual cleanup needed.

FTPS protocol notes: uPress requires explicit TLS on port 21 with `prot_c` (data channel protection). Plain FTP and implicit FTPS (port 990) do not work. The protocol is documented in the lean-kit `UPRESS_FTPS_PROTOCOL_v1.0.0.md`.

Post-deploy smoke checks: 1. `curl -sI https://sfa.nimrod.bio/ | head -1` → HTTP 200 2. Load `https://sfa.nimrod.bio/` — home renders with module grid. 3. Spot-check a crop page (`/crop-book/<slug>`) and `/market/`.

When to run: after any merge to main that changes `sfa_delivery/`, `public_assets/`, or PHP templates. Not automated; runs manually from waldhomeserver.

2.2 Data Push (Crop Book and Market Index)

What is pushed: rows from Postgres tables (`crops` , `crop_varieties` , `products` , `product_prices` , `crop_field_enrichment` , `crop_attribute`) to the delivery-tier MySQL.

Transport: HTTPS POST to `https://sfa.nimrod.bio/api/v1/ingest` (HMAC-authenticated). No IP restriction — Cloudflare accepts connections from any IP.

Script:

```
python -m organic_market_agent.publisher.sfa_ingest_push [--table TABLE] [--dry-run] [--limit N]
```

Two cadences: - **Price index (OMA market):** waldhomeserver cron at 06:30 daily. The scraper → normalizer → aggregator → publisher pipeline runs on waldhomeserver; after publish, `sfa_ingest_push.py` pushes the `products` and `product_prices` tables to MySQL. - **Crop book:** pushed from the Mac on change (not a daily cron). When Nimrod or agents enrich crop data (imports, validation console, enrichment runner), the operator runs `sfa_ingest_push.py --table crops` (or `all`) from the Mac or waldhomeserver.

Why data can originate from the Mac: HTTPS to Cloudflare does not require an allowlisted source IP. The HMAC signature provides the authentication guarantee. The Mac pushes crop-book data directly; waldhomeserver pushes price-index data via its daily cron.

3. First-Deploy Migration Runner

Before data can be pushed to a fresh uPress instance, the MySQL schema must be created. The migration runner is a web-accessible PHP script (`sfa_delivery/migrate.php`) that applies all pending numbered SQL migrations.

URL: `https://sfa.nimrod.bio/admin/migrate?token=<ADMIN_MIGRATE_TOKEN>`

Security: single-use bearer token in the query string, compared server-side with the `ADMIN_MIGRATE_TOKEN` env var (constant-time comparison). After migrations are applied, the token should be rotated or the `migrate.php` file removed from the deployed tree.

Migrations applied in order: `001_schema_migrations.sql` → `002_crops.sql` → `003_products.sql` → `004_crop_field_enrichment.sql` → `005_crop_attribute.sql`.

Each migration's version string is recorded in `schema_migrations` after successful application; re-running the runner skips already-applied versions.

4. Cron and Automation

4.1 waldhomeserver Cron Jobs

Schedule	Command	Purpose
06:30 daily		

Schedule	Command	Purpose
	<code>python -m organic_market_agent.scheduler.runner</code>	Market pipeline: collect → normalize → aggregate → publish → ingest push (products + prices)
Nightly (time TBD)	Reconciler (<code>reconciler.py</code> WP-A)	Postgres ↔ MySQL drift audit; queues corrective re-push if needed
Daily	<code>pg_dump</code> to <code>/data/backups/</code>	Local Postgres backup (30-day retention; same-disk only — see §7.2 debt)

The scheduler runner (`scheduler/runner.py`) self-gates: it runs every minute via system cron but executes the pipeline only if `is_enabled=True` in the `scheduler_config` table AND the current time matches `run_hour:run_minute` (± 1 minute). An overlap guard prevents concurrent runs.

4.2 Freshness Guard

`freshness_days` on the `products` table is computed at push time by the publisher (`today - last_price_date`). The delivery-tier index query filters by `freshness_days <= 7` to show only "fresh this week" products. If a source goes down and prices are not scraped for >7 days, its products automatically drop from the index — no manual intervention needed.

4.3 Development Workstation Scheduler Policy

The Mac dev workstation must NOT have an automated daily scheduler for the market pipeline. Simultaneous runs from Mac + server against the same DB would corrupt pipeline run tracking. The policy is documented in `documentation/05-admin-and-operations/DEVELOPMENT_WORKSTATION_SCHEDULER_POLICY.md` : on the Mac, all ingestion runs are manual (via the Flask admin UI or CLI) and on-demand only.

5. Security Model

5.1 HMAC-SHA256 Ingest Authentication

Every `POST /api/v1/ingest` request carries `X-SFA-Auth: sha256=<hmac_hex>` . The HMAC is computed over the raw request body bytes using the shared secret `SFA_INGEST_HMAC_SECRET` (32 bytes, base64-encoded, from `openssl rand -base64 32`).

On the delivery tier, `HmacAuthMiddleware.php` verifies the signature using PHP's `hash_equals()` (constant-time comparison, prevents timing attacks). Mismatch → HTTP 401; no body processing.

The secret is identical on waldhomeserver (in the backend `.env`) and on uPress (in the delivery `.env`). Rotation policy: annually or immediately on suspected leak. Rotation requires updating both env files and confirming the push pipeline still works.

5.2 FTPS IP Allowlist

uPress restricts outbound FTPS (port 21) to an IP allowlist. Registered entries: - waldhomeserver: 46.235.231.114 (static) - Mac dev: registered separately (may be a dynamic Bezeq IP; if changed, must be updated at uPress firewall)

This allowlist is enforced at the uPress network level, not in application code. It cannot be bypassed with software changes on our side.

5.3 FTPS Explicit TLS (`prot_c`)

The uPress FTPS server requires explicit TLS on port 21 with data channel protection (`PROT P` / `prot_c` in lftp). Control channel and data channel are both encrypted. Plain FTP or implicit FTPS (port 990) are not accepted.

5.4 Delivery-Tier `.env` Security

The `.env` file on uPress contains database credentials and the HMAC secret. Security hardening: - `chmod 600` on `.env` (PHP-FPM user reads; nginx cannot serve it). - `.htaccess` blocks direct HTTP access to `.env` (returns 403). - Secrets never committed to git. `.env.example` lists required keys without values. - No KMS available on shared hosting — `.env` is the only secret storage available.

5.5 Security Blocks in `.htaccess`

The delivery-tier `.htaccess` returns HTTP 403 for direct HTTP access to: `/.env`, `/composer.{json,lock}`, `/migrations/`, `/app/`, `/tests/`, `/vendor/`, `/logs/`

All HTTP is forced to HTTPS by an `.htaccess` rewrite rule (defense-in-depth; Cloudflare also enforces HTTPS).

5.6 Ingest Idempotency (Anti-Replay)

The `idempotency_key` in every ingest payload prevents double-application of the same push. If the same key arrives twice, the controller returns `{"duplicate": true}` with HTTP 200 without re-applying rows. This protects against transient network retries replaying the same payload.

The `ingest_log` table retains keys for 30 days (opportunistic pruning: ~1% of requests run a `DELETE ... WHERE applied_at < NOW() - INTERVAL 30 DAY`).

5.7 S003 Read-Only Delivery (Current)

In the S003 scope (current production), the delivery tier has no user write paths. All writes flow only through the HMAC-signed publisher push. There is no user authentication, no user data in MySQL, no community features. This minimizes attack surface.

S004 will introduce JWT-gated user writes (calculator saves, favorites). That will be a separate decision artifact with its own security design.

6. Docker Port Canon (Shared Workstation)

The Mac is a shared development workstation running multiple Docker-based projects. Port assignments are canonical and must not conflict:

Port	Container / Service	Project
5433	oma-postgres (PostgreSQL)	SmallFarmsAgents
5001	Admin Flask server	SmallFarmsAgents
8081	Public viewer (static HTML/JSON)	SmallFarmsAgents
5432	PostgreSQL	Other projects (not SFA)
8080	Frontend	TikTrack (not SFA)

Never use ports 5432 (conflicts with other Postgres), 8080 (TikTrack frontend), or 3306 (MySQL default — uPress MySQL is not exposed locally).

Docker compose file for the local Postgres: `docker-compose.yml` in the repo root. The service is named `oma-postgres` ; connection string: `postgresql://postgres:postgres@localhost:5433/organic_market_agent` .

7. Operational Constraints and Known Debt

7.1 Crop Book Data Push: Manual, Not Cron

The crop book data push is intentionally NOT a daily cron job. Crop data changes are episodic — re-pushing identical artifacts daily would be wasteful and would generate `ingest_log` churn. The operator runs `sfa_ingest_push.py` manually after each enrichment or import session.

Tracking item: `documentation/KNOWN_DEBT.md` §A.2 — "Daily cron auto-publish for crop book."
Promoted when a daily-changing crop data source (e.g. live market prices integrated into the crop book) is introduced.

7.2 Backup Posture (Risk — flagged 2026-05-30)

The home server takes daily local `pg_dump` backups to `/data/backups/` with 30-day retention. However: - **Same disk as data** — not offsite. - **No restore verification** — backups are untested. - **Blast-radius mitigation already in place:** the price index is reconstructible by re-scraping; the crop book's canonical source is the git repo (re-seedable). A home-server total loss would not lose authored data — only running state.

Recommended follow-up (not blocking): add offsite copy of price-index DB dump + periodic restore test. Tracked in `KNOWN_DEBT.md` as a server-infra item.

7.3 uPress MySQL Backups

uPress takes automatic nightly backups of the entire site as part of their service. For belt-and-suspenders, a `mysqldump` periodic export via SSH cron from waldhomeserver is deferred to a post-cutover ops WP.

7.4 Code Deploy Is Not Automated

There is currently no CI/CD pipeline for the `sfa_delivery/` code deploy. Merges to main do not automatically trigger an FTPS deploy. The operator runs `bash scripts/ftp_deploy_sfa_ui.sh` manually from waldhomeserver after each significant merge.

7.5 PHP Migration Runner Security

The `migrate.php` web-accessible migration runner (`/admin/migrate?token=...`) is intended for first-deploy use only. It should be rotated to an unguessable token (or the file removed) after migrations are applied. This is a known low-severity operational discipline item.

7.6 Waldhomeserver Home Server Reliability

waldhomeserver is a shared home server, not a managed cloud instance. It has: - No SLA or uptime guarantee. - No automatic restart of the daily cron if the machine is rebooted. - Residential internet connection (Bezeq); periods of outage are expected.

Impact on users: if waldhomeserver is down, price-index data on `sfa.nimrod.bio` becomes stale after 7 days (`freshness_days > 7`) and products drop from the market index. The crop book is unaffected (it is near-static data pushed from the Mac; the delivery tier serves it from MySQL with no runtime dependency on waldhomeserver).

7.7 Mobile Responsive Tuning (KNOWN_DEBT.md §A.3)

The crop book SPA was originally built desktop-first. Mobile parity is best-effort. The UX/UI overhaul WP (WP-B, team_35) explicitly addresses mobile-first design. This is a MEDIUM severity item expected to be consumed by WP-B.

7.8 Credentials Management

There is no centralized secrets manager or vault. All secrets live in `.env` files on each machine: - Mac: `.env` in repo root (gitignored). - waldhomeserver: `.env` in the deployment directory. - uPress: `.env` in the site root (FTPS-uploaded; `chmod 600`).

Rotation checklist documented in `documentation/05-admin-and-operations/FTPS_CRED_ROTATION_SYNC_RUNBOOK.md`. The HMAC secret must be identical on waldhomeserver and uPress and rotated in lockstep.

8. Troubleshooting Quick Reference

8.1 Pipeline Alerts

All pipeline alerts are persisted in the `pipeline_alerts` table (Postgres, walddhomeserver). Access via Flask admin at `/alerts` or SQL export. Alert prefix taxonomy:

Prefix	Meaning
<code>[OPS:process_restart]</code> / <code>[OPS:admin_stop_all]</code>	Expected lifecycle events after admin restart / stop-all
<code>[PIPELINE:failure]</code> / <code>[PIPELINE:missing_run]</code>	Real worker errors
<code>[SCHEDULER:...]</code>	Cron runner events (overlap guard, time gate, disabled)
<code>[SIMULATION:test]</code>	Pytest or <code>triggered_by=test</code> runs — filter after review
<code>[MAINTENANCE:...]</code>	Background maintenance finished or failed (renormalize, refresh)
<code>[AGG_PRICE_RULE:two_source_price_spread_gt_100pct]</code>	Publish suppressed: 2-source spread >100%
<code>[AGG_PRICE_RULE:multi_source_outlier_gt_2sigma]</code>	Publish suppressed: 3+-source outlier >2 σ

8.2 Common Operational Scenarios

Symptom	Likely cause	Resolution
Market prices stale on <code>sfa.nimrod.bio</code>	walddhomeserver cron not running	Check <code>scheduler_config.is_enabled</code> ; restart scheduler
New products not appearing	Below publish threshold (<2 observations, <2 sources)	Add aliases; check scope-skip rules
Crop book data stale after schema migration	New fields not yet pushed	Run <code>sfa_ingest_push.py --table crops</code> from Mac
<code>/api/v1/health</code> returns <code>db: fail</code>	MySQL connection issue on uPress	Check uPress DB status; check <code>.env</code> credentials
Ingest push returns 401	HMAC secret mismatch	Verify <code>SFA_INGEST_HMAC_SECRET</code> matches on both ends
Ingest push returns 200 <code>{duplicate: true}</code>	Same idempotency key reused (expected)	Normal on retry; no action needed
Admin shows old data	Stale Flask server process	<code>bash scripts/restart_all_servers.sh</code> <code>restart</code>
<code>alembic current</code> not at head	Missing migrations	<code>python3 -m alembic upgrade head</code>
Tests fail with <code>OperationalError</code>	PostgreSQL container not running	<code>docker-compose up -d</code>
FTPS deploy fails		

Symptom	Likely cause	Resolution
	Egress IP not allowlisted or wrong credentials	Verify waldhomeserver IP; check uPress allowlist; check <code>.env</code> FTPS creds

8.3 Server Management Commands

```
# Flask admin server (port 5001) – waldhomeserver or Mac
bash scripts/admin_server.sh start|stop|restart|status

# Public viewer (port 8081) – local static HTML server for dev
bash scripts/viewer_server.sh start|stop|restart|status

# Both servers
bash scripts/restart_all_servers.sh restart

# Database health check
python -m organic_market_agent.db.check

# Push all tables to delivery tier (dry run first)
python -m organic_market_agent.publisher.sfa_ingest_push --dry-run
python -m organic_market_agent.publisher.sfa_ingest_push --table all

# UI code deploy (from waldhomeserver)
bash scripts/ftp_deploy_sfa_ui.sh
```

9. Environment Variables Reference

Variable	Where used	Purpose
<code>DATABASE_URL</code>	Python (all)	PostgreSQL connection string (<code>postgresql://...@localhost:5433/...</code>)
<code>SFA_INGEST_URL</code>	<code>sfa_ingest_push.py</code>	Target ingest endpoint (<code>https://sfa.nimrod.bio/api/v1/ingest</code>)
<code>SFA_INGEST_HMAC_SECRET</code>	<code>sfa_ingest_push.py</code> + <code>HmacAuthMiddleware.php</code>	32-byte base64 HMAC secret (must match both ends)
<code>ADMIN_SECRET_KEY</code>	Flask admin	Session signing key
<code>ADMIN_MIGRATE_TOKEN</code>	<code>migrate.php</code> (uPress)	One-time migration runner auth token
<code>SFA_FTP_HOST</code>	<code>ftp_deploy_sfa_ui.sh</code>	uPress FTPS hostname
<code>SFA_FTP_PORT</code>	<code>ftp_deploy_sfa_ui.sh</code>	21 (explicit TLS)
<code>SFA_FTP_USER</code> / <code>SFA_FTP_PASS</code>	<code>ftp_deploy_sfa_ui.sh</code>	uPress FTPS credentials
<code>SFA_FTP_R00T</code>	<code>ftp_deploy_sfa_ui.sh</code>	FTPS target path on uPress (typically <code>/</code>)
<code>SFA_DB_NAME</code> (uPress PHP <code>.env</code>)	<code>IngestController.php</code> / PDO	MySQL database name (<code>sfanms2u_SFAUserUiDB</code>)
	PDO	MySQL credentials

Variable	Where used	Purpose
SFA_DB_USER / SFA_DB_PASS (uPress PHP .env)		
NORMALIZER_BASELINE_JSON	Flask admin dashboard	Optional baseline path for delta display
RAW_FILES_ROOT	Collectors	Raw asset storage directory
AOS_ACTOR_API_KEY	AOS messaging	Actor key for AOS hub API (see reference_aos_actor_key.md)

10. Deployment Lifecycle Checklist

For each release that changes the delivery tier:

- [] Code changes merged to main in `sfa_delivery/` or `public_assets/`
- [] If schema changed: new numbered migration SQL file added (`sfa_delivery/migrations/XXX_*.sql`)
- [] `composer.lock` updated if new PHP deps added
- [] `vendor/` NOT committed (gitignored; deploy script runs `composer install`)
- [] Deploy from waldhomeserver: `bash scripts/ftp_deploy_sfa_ui.sh`
- [] If schema migration: run `https://sfa.nimrod.bio/admin/migrate?token=<TOKEN>` once
- [] Post-deploy smoke: `curl HTTP 200`; load site; spot-check crop page and market page
- [] If crop data changed:
 - `python -m organic_market_agent.publisher.sfa_ingest_push --table crops`
- [] Monitor `/api/v1/health` for db: ok

For rollback: re-deploy the previous known-good commit's `sfa_delivery/` tree (lftp `--delete` makes it idempotent).

11. Future S004 Infrastructure Considerations

When opening S004 (calculator persistence + community features), the following infrastructure changes are anticipated:

- **JWT authentication** — a 50-line JWT helper is already designed (deferred to S004). Routes for user writes will be separate from the current read-only surface.
- **User tables in MySQL** — `user_favorites`, `user_calculator_inputs`, per-user state. Separate decision and migrations from current S003 schema.
- **Community submission tables** — comments, crop corrections. Separate decision; separate auth scheme.
- **Potential host migration** — if uPress shared hosting becomes a constraint (user writes, persistent sessions, connection pools), migration to a managed LAMP host (DigitalOcean, Hetzner, A2) is straightforward given the portability-first architecture (§12 in doc 06). No uPress-specific bindings exist.

Pre-S004 operational items to resolve (from `documentation/KNOWN_DEBT.md`): AOS actor API key provisioning (B.1); GCR status updates (B.2); mobile responsive debt (A.3); Tend cross-year ingestion for calculator data (C.1).

Document 07 — authored 2026-06-03 by team_100 for the SFA Product Information Pack. Sources: sfa-delivery-tier.md; UI_DEPLOY_RUNBOOK.md; FTPS_CRED_ROTATION_SYNC_RUNBOOK.md; 08-troubleshooting/README.md; KNOWN_DEBT.md; 05-admin-and-operations/README.md; sfa_ingest_push.py.